IT3091 Machine Learning - Personal Learning Journey

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My main responsibility was the vegetable wholesale-price analysis and preparation stage. The project's price question is specifically about the next observed weekly Colombo-and-suburbs wholesale price, so keeping the source frequency weekly was an important decision for me.

The weekly workbook stores prices across W1 to W52 columns. I learned to inspect the workbook structure, filter for wholesale observations, reshape the week columns into crop-week records, and align commodity labels with the crop names used elsewhere in the project. This produced 15,598 prepared weekly price observations across 26 crops in the deployed coverage view.

For modelling, records need both an available previous price and a next observed price target. After those requirements were applied, 15,546 weekly rows were used by the price model. The lag and target are based on observed records; if a source week is missing, the next observation can be more than seven days later. This is why I learned not to fill missing weeks without evidence.

My work supported the price model that uses crop, market, calendar fields, the current price, and the previous observed price to estimate the next one. The app also lets a user select a crop to inspect its historical trend and filter the recent data-coverage table. The stored series ends on 3 March 2025, so the forecast page asks users to enter current market values for a current estimate.

This stage improved how I think about data quality: workbook labels, units, frequency, and crop mapping all affect the final prediction. If I continued, I would add verified market drivers such as supply, demand, transport, or festival timing and validate forecasts on later unseen weeks. I learned that a reproducible weekly table is part of the model, not just a preliminary cleaning step.